

Pablo Leyva

908-525-6062 | pl33@njit.edu | linkedin.com/in/pablo-leyva | github.com/pleyva2004 | pabloleyva.io

Dear Google AI Career Catalyst Recruiting Team,

I am a senior at the New Jersey Institute of Technology completing a B.S. in Applied Mathematics and Computer Science (AI concentration, May 2027), and I am applying to the AI Career Catalyst Program because its premise, engineering at the intersection of AI and infrastructure, is exactly the intersection where I have spent the last three years building.

On the AI side, I am currently in my second internship at Apple, where I build model-based tools that curate and score RL post-training data for a continually learning small language model, and where I am implementing GRPO/PPO on-policy training with vLLM rollouts against verifiable-reward environments. Last summer at Apple I built an agent orchestration system on the Apple Foundation Model with MCP and the team's LLM evaluation pipeline. At NJIT, my research fellowship has me training a 3B-parameter reasoning model with multi-GPU distributed training (FSDP) through an SFT, DPO, and RL fine-tuning pipeline.

On the infrastructure side, I built NeuroCache, an open-source KV-cache memory manager with attention-scored eviction and cache-replacement policies, and at Caterpillar I built an Airflow pipeline ingesting 500K+ daily records with FastAPI model serving at sub-100ms latency. The models I care about only matter when the systems underneath them are efficient and reliable, and that is the layer I want to master at Google's scale.

The rotational format is why I am drawn to this program specifically: I have proven I ramp quickly in unfamiliar territory. I have interned across heavy industry and consumer technology, and with a team of five engineers at SHPE NJIT I built and shipped our chapter's first iOS app in 27 days, growing it to 200+ users in its first month. As SHPE chapter president I lead a 20-person board serving 300+ students. I grew up in Mexico City in a family that treated education as the way forward, and I bring that same conviction, and the responsibility to open doors for others, to every team I join.

I graduate in May 2027 and am available to start in June 2027. I would welcome the chance to speak further about the program. Thank you for your consideration.

Sincerely,

Pablo Leyva